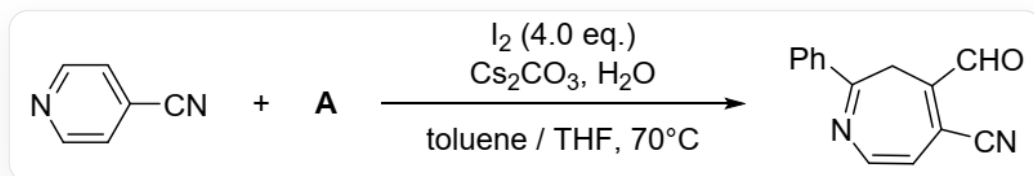


Question

People utilized the following reaction to construct the azepine ring system:



In a mixed solvent of toluene and THF, at 70 degrees Celsius, N#CC1=CC=NC=C1 and **A** react with Cs2CO3, H2O, and 4 equivalents of iodine to yield O=CC1=C(C#N)C=CN=C(C2=CC=CC=C2)C1

Mechanism studies indicate that the reaction proceeds through the key intermediate $\mathbf{X}^+(C_{14}H_{12}IN_2)$; the salt of $\mathbf{X}^+$ can be isolated even if CsCO3 and H2O are not added to the reaction. $\mathbf{X}^+$ further reacts with I2 under the above reaction conditions to generate the final product.

Choose the correct option from the following:

- A.** All other options are incorrect
- B.** The molecular weight of **A** is approximately 92.
- C.** The degree of unsaturation of **A** is 4.
- D.** $\mathbf{X}^+$ contains only 1 hexagon
- E.** $\mathbf{X}^+$ to obtain the product also requires consuming 1 molecule of iodine and 4 molecules of cesium carbonate (considered as a monobasic base)

F. To obtain `COC(=O)C1=CN=C(CC(=C1)C=O)C2=CC=C3C=CC=CC3=C2` under the same conditions, the two required reactants are `O=C(OC)C1=CC=NC=C1` and `C=CC1=CC2=CC=CC=C2C=C1`

Answer

Correct Answer: **E**

Detailed Explanation

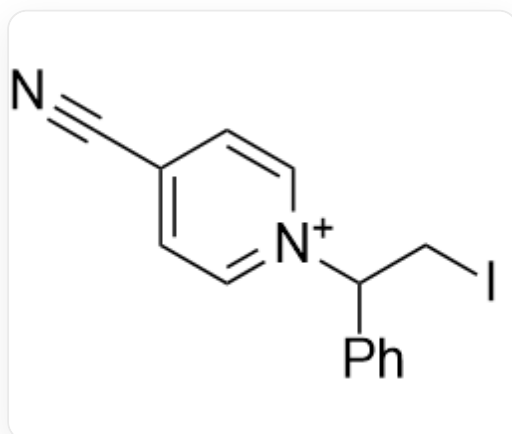
$\mathbf{X}^+$ has C_8H_8 and one iodine more than N#CC1=CC=NC=C1, so the molecular formula of **A** should be C_8H_8 . The most reasonable C_8H_8 alkene is styrene. Therefore, the molecular weight of **A** is 104 and the degree of unsaturation is 5.

CHECKPOINT

2 PTS

The molecular formula of **A** should be C_8H_8 , the molecular weight is 104 and the degree of unsaturation is 5, options BC are incorrect

After the double bond in **A** reacts with iodine, it is nucleophilically attacked by N#CC1=CC=NC=C1 to obtain $\mathbf{X}^+$: ICC(C1=CC=CC=C1)[N+]2=CC=C(C#N)C=C2, which has two six-membered rings.



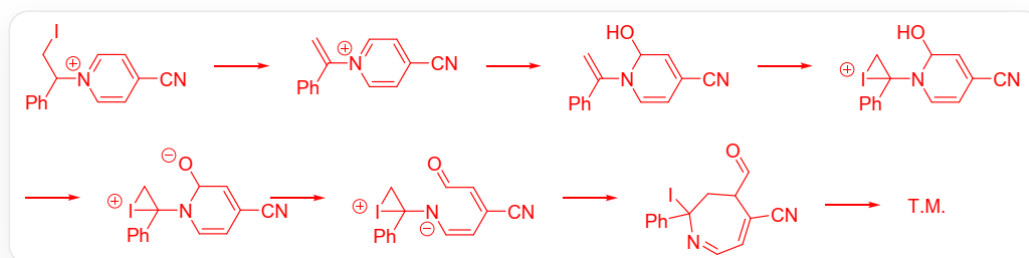
ICC(C1=CC=CC=C1)[N+]2=CC=C(C#N)C=C2

CHECKPOINT

1 PTS

The structure of $\mathbf{X}^+$ is ICC(C1=CC=CC=C1)[N+]2=CC=C(C#N)C=C2, which has two six-membered rings, option D is incorrect

The mechanism for obtaining the product from $\mathbf{X}^+$ is:



First, cesium carbonate, as a base, eliminates *HI* in **X**⁺ to generate a double bond: `C=C(C1=CC=CC=C1)[N+]2=CC=C(C#N)C=C2`, then a water molecule attacks and departs a proton to generate a hydroxyl group: `C=C(N1C=CC(C#N)=CC1O)C2=CC=CC=C2`, the newly generated double bond is attacked by a molecule of new iodine: `N#CC1=CC(O)N(C2(C3=CC=CC=C3)C[I+]2)C=C1`, the hydrogen of the newly generated hydroxyl group is abstracted by a base: `N#CC1=CC([O-])N(C2(C3=CC=CC=C3)C[I+]2)C=C1`, then the six-membered ring opens: `N#CC(/C=C/[N-]C1(C2=CC=CC=C2)C[I+]1)=C/C=O`, the negative charge on the nitrogen attacks along the conjugated double bond to open the iodine three-membered ring to generate a seven-membered ring: `N#CC1=CC=NC(I)(C2=CC=CC=C2)CC1C=O`, finally the base abstracts the α-hydrogen of the aldehyde group with the strongest acidity, and the iodine atom leaves to obtain the product

According to the mechanism, one molecule of iodine is consumed and 4 protons need to be consumed. Each molecule of cesium carbonate can neutralize 1 proton (cesium bicarbonate is not basic enough), so 1 molecule of iodine and 4 molecules of base need to be consumed.

CHECKPOINT

1 PTS

Obtaining the product from $\mathbf{X}^+$ requires consuming 1 molecule of iodine and 4 molecules of base, option E is correct

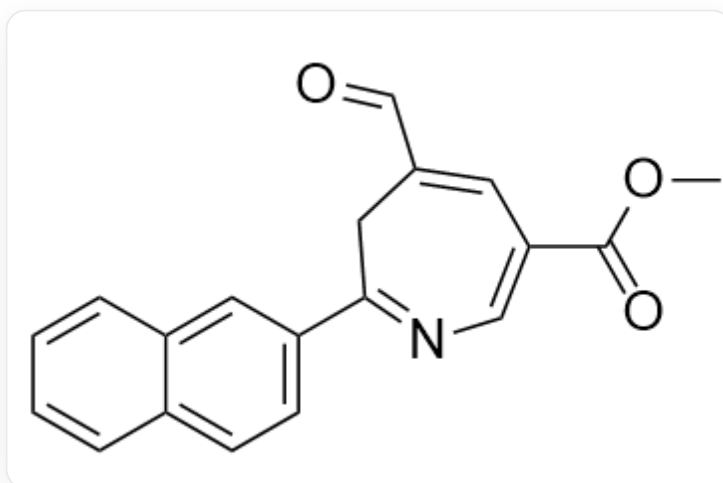
According to the mechanism, the aryl group on the alkene is located at the 2-position, the newly generated aldehyde group is located at the 4-position, the two 2-positions of the original pyridine are the 7-position and the aldehyde carbon of the seven-membered ring, respectively, the two 3-positions of the pyridine are the 6-position and the 4-position of the seven-membered ring, respectively, and the 4-position of the pyridine is the 5-position of the seven-membered ring. The aryl group of `COC(=O)C1=CN=C(CC(=C1)C=O)C2=CC=C3C=CC=CC3=C2` is a naphthyl group, so the corresponding alkene in the reaction is `C=CC1=CC2=CC=CC=C2C=C1`; the substituent methyl ester group is located at the 6-position of the seven-membered ring, corresponding to the 3-position of the original pyridine, and the pyridine derivative is `O=C(OC)C1=CC=CN=C1`.

CHECKPOINT

1 PTS

The two reactants are `C=CC1=CC2=CC=CC=C2C=C1` and `O=C(OC)C1=CC=CN=C1` respectively, option F is incorrect

The structure of `COC(=O)C1=CN=C(CC(=C1)C=O)C2=CC=C3C=CC=CC3=C2` is:



`COC(=O)C1=CN=C(CC(=C1)C=O)C2=CC=C3C=CC=CC3=C2`